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Baicalin confers neuroprotection in animal models of stroke through its antioxidant and anti-apoptotic effects

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ABSTRACT

Importance: Ischemic stroke leads to neuronal cell death due to a lack of oxygen and glucose. Baicalin is a flavonoid that has antioxidant and anti-inflammatory properties.

Objective: The aim of this study is to elucidate the anti-oxidant and anti-apoptotic effects of baicalin in animal models of stroke.

Methods: Vehicle or baicalin (100 mg/kg) was administered intraperitoneally immediately after the middle cerebral artery occlusion (MCAO) surgery. Neurobehavioral tests were conducted 24 h post-MCAO and brain tissue was isolated to assess histopathological changes and apoptosis-associated protein expression. Additionally, reactive oxygen species (ROS) and lipid peroxidation (LPO) assays were performed to evaluate oxidative stress.

Results: MCAO animals exhibited severe neurological deficits, which were significantly alleviated by baicalin treatment. Baicalin mitigated the up-regulation in ROS and LPO levels induced by surgery. MCAO damage led to severe histopathological lesions and an increase in terminal deoxynucleotidyl transferase dUTP nick end labeling-positive reactions, these alterations were alleviated by baicalin treatment. MCAO damage decreases the expression of Bcl-2 and increases the expression of Bax, baicalin alleviates these changes. Baicalin also attenuated the upregulation of caspase-3 expression caused by MCAO injury.

Conclusions and Relevance: These results can suggest evidence that baicalin exerts neuroprotective effects by preventing apoptosis during cerebral ischemia. In conclusion, baicalin acts as a potent neuroprotective agent through its antioxidant and anti-apoptotic effects on neuronal cell damage.

Keywords: Baicalin; cerebral ischemia; neuroprotection; stroke

INTRODUCTION

Stroke is the most prevalent cerebrovascular disease and remains a leading cause of death and disability worldwide [1]. Stroke is classified into ischemic and hemorrhagic types [2]. The major symptoms of ischemic stroke include loss of balance, dizziness, and blurred weakness of the arms and legs. Ischemic stroke due to vascular occlusion results in a decrease

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Conflict of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The datasets used and/or analyzed during the current study are available from the corresponding author upon reasonable request.

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in cerebral blood flow and leads to oxygen deprivation [3]. This causes ion pump dysfunction and oxidative damage, resulting in damage to brain tissue within a few hours [3]. Cerebral ischemic injury disrupts the blood-brain barrier (BBB), causing permanent neurological deficits [4]. It also increases intracellular calcium and harmful molecules, induces oxidative stress and DNA damage, and activates apoptotic pathways [5,6]. As described above, the mechanisms underlying cerebral ischemic injury are complex and multifaceted, making it difficult to achieve effective recovery of brain tissue following ischemic injury.

Baicalin is a flavonoid compound derived from the roots of *Scutellaria baicalensis* and *Scutellaria lateriflora*, both of which are key components in traditional Chinese herbal medicine and Japanese Kampo medicine [7]. Baicalin exhibits anti-inflammatory, anti-oxidant, anti-cancer, and anti-viral properties [8-10]. Baicalin also exerts potent antidepressant effects by suppressing neuronal apoptosis in models of chronic and unpredictable mild stress [11]. Baicalin promotes the neuronal differentiation of neural stem cells and protects neurons against cerebral ischemia-reperfusion injury [12,13]. Furthermore, baicalin reduces BBB permeability during hypoxia and attenuates the inflammatory response following ischemic neuronal injury [9,10]. Although the benefits of baicalin have been reported, further studies are required to elucidate the protective effects of baicalin on cerebral ischemia. The aim of this study was to evaluate the neuroprotective effects of baicalin in animal models of stroke through anti-oxidation and apoptosis suppression. By investigating the effects of baicalin on neuronal survival, this study provides insight into the potential therapeutic benefits of baicalin in the treatment of stroke. We performed various neurological behavioral tests and histopathological evaluations for this study. We also investigated oxidative stress levels and assessed changes in typical apoptotic regulatory proteins Bcl-2, Bax, and caspase-3.

METHODS

Experimental animals

Sprague-Dawley rats (male, 7 weeks old, 210–230 g, n = 48; Samtako Co, Korea) was used as experimental animals. Animals were provided free to food and water, and were maintained in animal rooms where light (12-h light/dark) and temperature (25°C) were controlled. The animals were grouped as follows: vehicle + sham, baicalin + sham, vehicle + middle cerebral artery occlusion (MCAO), and baicalin + MCAO. Baicalin (Sigma-Aldrich, USA) was freshly dissolved in phosphate-buffered saline (PBS) immediately before administration. The solution was visually checked to confirm that no sediment had formed. Baicalin was administered intraperitoneally at a dose of 100 mg/kg immediately following MCAO surgery. We have determined the dosage of baicalin treatment according to a previous study [14]. Vehicle indicates only PBS administration. Twelve rats per group were used for the following experiments: reactive oxygen species (ROS) and lipid peroxidation (LPO) assay (n = 4 for each group), morphological study (n=4 for each group), and Western blot analysis (n = 4 for each group). Neurobehavioral tests were conducted in all animals (n = 48). All experimental procedures were carried out according to the guideline of the Institutional Animal Care and Use Committee of Gyeongsang National University (GNU-240527-R0110).

Middle cerebral artery occlusion surgical procedure

The surgical procedure was performed according to the manual described previously [15]. Anesthesia was induced by intramuscular injection of Zoletil (50 mg/kg; Virbac, France). The right common carotid artery (CCA) was exposed and isolated from nearby nerves and

tissues. The right external carotid artery (ECA) and the right internal carotid artery (ICA) were continuously exposed. The ends of the pterygopalatine artery and occipital artery were tied separately with 6-0 black silks, and the middle of both ends was cut. The ends of the right ECA were bound with black silk and the right CCA was clamped with micro-clamp to temporarily block the blood flow. The right ECA was cut and the probe (Doccol, USA) was inserted into the cut right ECA and pushed towards the ICA until it was no longer advancing. The right ECA and inserted probe were tied with black silk and the micro-clamp was removed from the right CCA to resume the blood flow. As a final step, the incision was sutured with 3-0 black silk and disinfected with povidone-iodine. Neurobehavioral tests were carried out 24 h after occlusion and animal were immediately sacrificed.

Neurological deficit scoring test

The animals were grouped into five groups by the following criteria: 1) score 0, no neurological impairment with a normal posture; 2) score 1, incomplete extension of the forelimb on the left side; 3) score 2, involuntary circling to the left side; 4) score 3, falling to the left side, seizures; and 5) score 4, lack of movement or consciousness.

Corner test

Two boards ($30 \times 20 \times 1 \text{ cm}^3$) were placed at an angle of 30° and rats were placed in the space between the boards. As the rats moved towards the corner and turned to left or right. The number of turns was recorded in 10 trials and the interval for this test was 30 sec. Animals underwent a seven-day training period prior to MCAO surgery, in which animals with the same number of right and left turns were used for corner tests.

Grip strength test

We used grip strength meter (Jeungdo-Bio and Plant Co., Ltd., Korea) to measure grip strength. The animals were placed on a grip test device and gripped the wire grid of the device with their front paws. The animals were slowly pulled back until they let go of their grips and the measured maximum force was recorded. Three tests were performed on each animal, and the average value was considered as a result.

Vibrissae-evoked forelimb placing test

The evaluation of vibrissae-induced forelimb placement was performed to measure sensorimotor aptitude [16]. Animals were allowed to grip tightly around the torso and move freely on both legs. As described manual, the right or left vibrissae of animals were brushed on the edge of a table. Upon stimulation of the vibrissae, the rats placed exactly corresponding forelimbs. Since the vibrissae sensory input is processed by the ipsilateral somatosensory cortex and induces a forelimb placement response on the same side, damage to the corresponding hemisphere impairs both the sensory input and the ipsilateral forelimb response. Each animal underwent 10 trials and recorded the success rate of the forelimb placement response for both right and left forelimb responses. Animals were trained for this assessment prior to MCAO surgery.

ROS assay

The 2',7'-dichlorodihydrofluorescein diacetate (DCFH-DA) has been used to measure intracellular ROS. We used a ROS assay kit and processed according to manufacturer's manual (Elabscience Biotechnology Inc., USA). The tissue was finely cut, reacted with an enzyme solution at 37°C for 30 min, rinsed with PBS, and reacted with 5 mM DCFH-DA at 37°C for 1 h. After centrifuging at 1,000 g for 10 min, the pellets were rinsed with PBS

for 5 min and centrifuged again. Fluorescence intensity was measured using a Multiskan EX spectrophotometer (Thermo Fisher Scientific, USA) at excitation/emission maxima of 488/525 nm.

LPO analysis

Malondialdehyde (MDA) has been used to measure LPO levels in the cerebral cortex, it is known as a marker of membrane polyunsaturated fatty acid peroxidation [17]. We used a LPO colorimetric measurement kit (BioVision Inc., USA) and processed it according to the manufacturer's instructions. The tissues were homogenized with MDA lysis buffer and centrifuged at 13,000 g for 10 min. The supernatant was collected, mixed with thiobarbituric acid, reacted at 95°C for 1 h, and cooled at 4°C for 10 min. The optical density at 532 nm wavelength was measured by spectrophotometer. Results were expressed in relative MDA pmol/mg of the protein.

2,3,5-triphenyltetrazolium chloride (TTC) staining

Brain tissues were separated from the skull and were sectioned coronally at 2 mm intervals using a brain matrix (Doccol). These sections were reacted in TTC (Sigma-Aldrich) dissolved in a phosphate buffer solution at 37°C for 10 min. They were fixed in a 4% neutral buffered paraformaldehyde solution for 4 h, sequenced according to brain regions, and digitally scanned. Quantitative analysis of intact and damaged areas was performed using Image J software (National Institutes of Health, USA). The percentage of infarcts was calculated according to the following formula:

$$\text{Infarct Percentage} = (\text{Infarct Volume} / \text{Total Slice Volume}) \times 100.$$

Hematoxylin and eosin (H&E) staining

The tissue was immersed in a 4% neutral buffered paraformaldehyde solution for 48 h, rinsed with tap water, and dehydrated step by step in 70% to 100% ethanol. The tissue was replaced with paraffin (Leica, Germany) using the Leica EG1160 paraffin embedding center (Leica) and the paraffin block were cut using HistoCore AUTOCUT (Leica). These sections were stained according to the general manual of H&E staining. The stained tissue was covered with a cover glass and observed with a Leica microscope (Leica). Five fields of the cortical area were selected and the results were shown as a percentage of the number of damaged cells and the total number of cells.

Terminal deoxynucleotidyl transferase dUTP nick end labeling (TUNEL) assay

ApopTag peroxidase in situ apoptosis detection kit (Merck, USA) was used to detect apoptotic cells. The sections were reacted with proteinase K (20 µg/mL) and rinsed with PBS. They were immersed in 3% hydrogen peroxide diluted with methanol for 5 min to inhibit endogenous peroxidase activity. After washing with PBS, they were incubated with equilibration buffer at 4°C for 1 h, continuously incubated with terminal deoxynucleotidyl transferase enzyme and reaction buffer mixture at 37°C for 90 min, and covered with stop buffer for 10 min. They were reacted with anti-digoxigenin conjugate for 1 h, reacted with 3,3'-diaminobenzidine (DAB; Sigma-Aldrich), and rinsed with tap water. They were counter stained with hematoxylin solution and mounted with a permount mounting medium (Thermo Fisher Scientific). We selected five fields in the cerebral cortical regions and counted TUNEL positive cells.

Western blot analysis

The right cerebral cortex was carefully collected and homogenized with lysis buffer containing phenylmethanesulfonyl fluoride (Sigma-Aldrich). Lysates were sonicated for 30 sec and centrifuged at 15,000 g for 1 h at 4°C. The supernatants were carefully collected and the concentration of protein was measured using bicinchoninic acid protein assay kit (Pierce, USA). Total proteins (30 µg) from each sample were lysed with sodium dodecyl sulfate (SDS) buffer, heated at 100°C for 15 min, and cooled to room temperature. SDS polyacrylamide gel (12%) was used and sample was run at 10 mA for 90 min and transferred to a polyvinylidene fluoride membrane (GVS Filter Technology, USA) at 120 V for 2 h. The membranes were blocked with 5% skim milk solution in Tris-buffered saline containing 0.1% Tween-20 (TBST) for 1 h, washed 10 min with TBST, and reacted for overnight at 4°C with specific primary antibodies: anti-Bcl-2, anti-Bax, anti-caspase 3, and anti-β-actin (diluted 1:1,000; Cell Signaling Technology, USA). They were washed with TBST and were reacted with horseradish peroxidase-conjugated anti-rabbit IgG (diluted 1:5,000; Cell Signaling Technology) for 1 h. They were incubated with enhanced chemiluminescence reagents (GE Healthcare, UK). The image was measured using ChemiDoc Image System (Bio-Rad, USA) and the density of band using Image J software. Results showed a ratio of relative density of specific protein to density of β-actin.

Immunofluorescence staining

The deparaffinized sections were rinsed with PBS and incubated with 1% normal goat serum for 1 h. They were reacted primary antibodies: anti-Bcl-2, anti-Bax, and anti-caspase-3 rabbit IgG (diluted 1:100; Cell Signaling Technology). After incubation, they were washed with PBS and incubated with fluorescein isothiocyanate-conjugated secondary antibody (1:100 dilution; Thermo Fisher Scientific) for 1 h. The sections were washed with PBS, stained with 4',6-diamidino-2-phenylindole (DAPI; Sigma-Aldrich) for 10 min, mounted using a fluorescent mounting medium, and examined with a confocal microscope (FV-1000; Olympus). The results were expressed as a ratio of the number of caspase-3 positive cells to that of DAPI positive cells.

Data analysis

All experiment data were presented as the mean ± standard error of means. The results of each group were compared by two-way analysis of variance and *post-hoc* Scheffe's test. The *p* value less than 0.05 was regarded as statistically significant.

RESULTS

Alleviation of neurobehavioral disorder by baicalin administration in MCAO animals

Fig. 1A showed the chemical structure and molecular formula of baicalin. **Fig. 1B-E** showed that administration of baicalin improves neurological impairment in MCAO animals through various neurological behavioral test. Surgical damage caused severe neurological and functional deficits, which were significantly attenuated by baicalin treatment. MCAO animals showed following behavioral deficits: circling, contralateral falls, seizures, and reduced mobility. The neurological deficit scores were 3.36 ± 0.42 in the vehicle + MCAO animals and 2.21 ± 0.38 in the baicalin + MCAO animals (**Fig. 1B**). In the corner test, MCAO animals predominantly rotated to the right side, but this tendency was mitigated by baicalin administration. The number of right turns was 9.71 ± 0.32 in the vehicle + MCAO animals

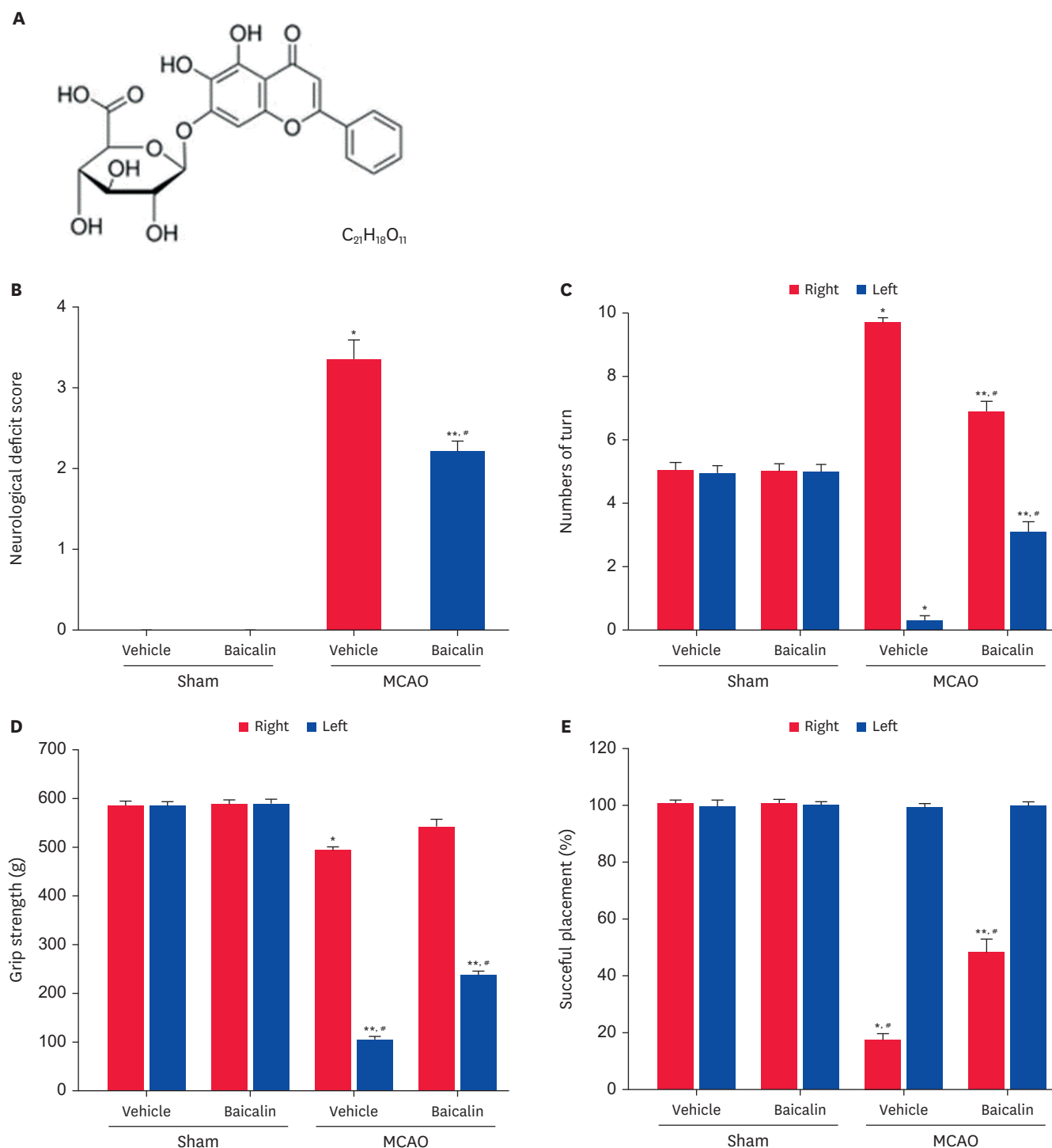


Fig. 1. Baicalin attenuates neurobehavioral deficits in MCAO animals. The chemical structure and molecular formula of baicalin (A), neurological deficit scoring test (B), corner test (C), grip strength assessment test (D), and vibrissae-evoked forelimb placing test (E) in the vehicle + sham, baicalin + sham, vehicle + MCAO, and baicalin + MCAO animals. Baicalin improved neurobehavioral disorder caused by MCAO damage. Data ($n = 10$) are shown as means $\pm$ standard error of means. MCAO, middle cerebral artery occlusion.

* $p < 0.01$, ** $p < 0.05$ vs. vehicle + sham animals; # $p < 0.05$ vs. vehicle + MCAO animals.

and 6.91 ± 0.53 in the baicalin + MCAO animals (**Fig. 1C**). Forelimb pulling force was assessed using the forelimb grip strength test. MCAO animals exhibited significantly lower grip strength scores compared to sham animals, baicalin treatment improved this reduction. Scores in sham animals remained consistent, irrespective of baicalin treatment. In the left forelimbs, grip strength was 105.25 ± 6.45 g in the vehicle + MCAO animals and 237.25 ± 8.34 g in the baicalin + MCAO animals. In the right forelimbs, grip strength was 494.12 ± 6.60 g in the vehicle + MCAO animals and 542.51 ± 14.79 g in the baicalin + MCAO animals (**Fig. 1D**). The vibrissae-evoked forelimb placement test assessed the animal's sensorimotor response capacity. MCAO injury caused severe sensorimotor impairment, baicalin ameliorated this disorder. Sensorimotor response scores were significantly reduced in MCAO animals compared to sham animals, with scores of 17.58 ± 2.05 in the vehicle + MCAO animal and 48.24 ± 4.69 in the baicalin + MCAO animal (**Fig. 1E**).

Antioxidative effect of baicalin administration in MCAO animals

ROS and LPO levels were quantified using 2',7'-dichlorodihydrofluorescein (DCF) and MDA assays to evaluate oxidative stress. MCAO animals exhibited elevated DCF levels, which were significantly reduced by baicalin treatment. The DCF levels was 3.65 ± 0.19 pmol/mg protein in the vehicle + MCAO animals and 2.63 ± 0.16 pmol/mg in the baicalin + MCAO animals (**Fig. 2A**). MDA levels was increased in MCAO animals, this elevation was mitigated by baicalin treatment. MDA levels was 3.18 ± 0.31 pmol/mg protein in the vehicle + MCAO animals and 2.46 ± 0.11 pmol/mg in the baicalin + MCAO animals (**Fig. 2B**).

Mitigation of infarction by baicalin administration in MCAO animals

A significant increase in infarct volume was observed in animals subjected to MCAO. However, baicalin administration attenuated the infarct volume compared to that in the vehicle-treated group (**Fig. 2C**). Sham-operated animals exhibited no infarct region. Quantitative analysis revealed infarct volumes of $38.4\% \pm 5.15\%$ and $21.7\% \pm 4.35\%$ in the vehicle + MCAO and baicalin + MCAO groups, respectively (**Fig. 2D**).

Mitigation of histopathological damage and apoptotic cells by baicalin administration in MCAO animals

Histopathological changes in MCAO animals were assessed using H&E staining (**Fig. 3A**). Sham animals had normal cell morphology characterized by large and round nuclei, distinct nucleoli, and dense cytoplasmic structures. However, MCAO animals exhibited damaged neurons characterized by contracted or distended cell shapes, nuclear condensation, and loss of nucleoli. Baicalin ameliorated MCAO-induced histopathological alterations. The percentage of damaged cells was $78.23\% \pm 5.75\%$ in vehicle + MCAO animals and $50.88\% \pm 6.92\%$ in baicalin + MCAO animals (**Fig. 3C**). Additionally, TUNEL histochemical staining was performed to detect apoptosis in the cerebral cortex (**Fig. 3B**). The number of TUNEL-positive cells was significantly higher in vehicle + MCAO animals compared to sham animals. However, these increases were significantly attenuated in baicalin + MCAO animals. The percentage of apoptotic cells was $90.15\% \pm 6.31\%$ in vehicle + MCAO animal and $49.69\% \pm 4.13\%$ in baicalin + MCAO animals (**Fig. 3C**).

Change of Bcl-2, Bax, and caspase-3 expression by baicalin administration in the MCAO animals

We investigated the expression of Bcl-2, Bax, and caspase-3 (**Figs. 4 and 5**). A decrease in Bcl-2 and an increase in Bax were observed after MCAO surgery, and these changes were alleviated by baicalin administration (**Fig. 4A**). Western blot analysis revealed that the level of Bcl-2

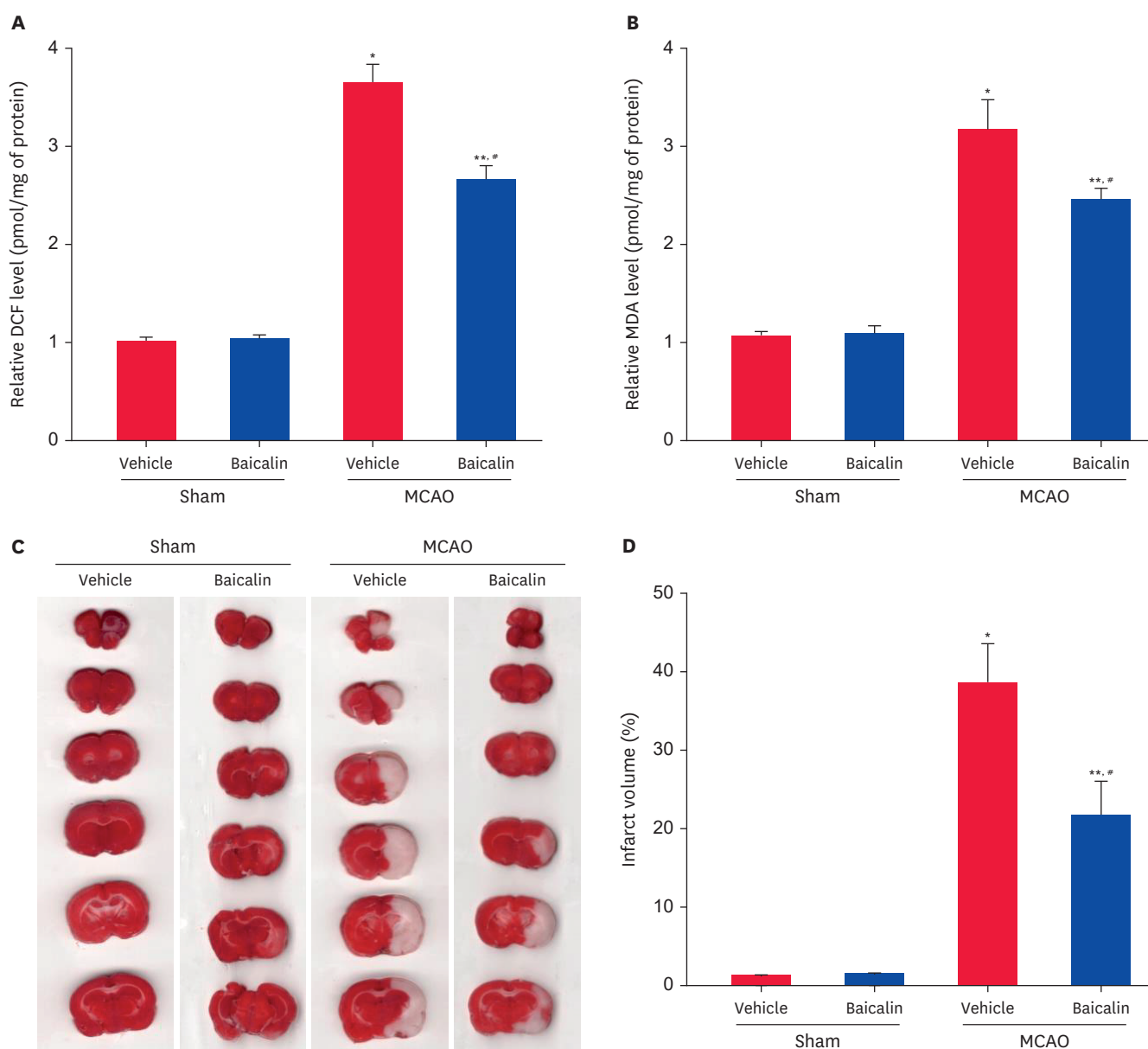


Fig. 2. Baicalin exerts a neuroprotective effect in MCAO animals. ROS (A) and LPO (B) assays, representative microphotograph of triphenyltetrazolium chloride staining (C and D) of the brain in the vehicle + sham, baicalin + sham, vehicle + MCAO, and baicalin + MCAO animals. DCF and MDA levels were increased in vehicle + MCAO animals, while baicalin treatment attenuated these increases. Intact areas were stained red color, while ischemic areas were remained white color. Baicalin improved the infarct volume increased by MCAO damage. Data (n = 4) are shown as means $\pm$ standard error of means. ROS, reactive oxygen species; LPO, lipid peroxidation; MCAO, middle cerebral artery occlusion; DCF, 2',7'-dichlorodihydrofluorescein; MDA, malondialdehyde. * $p < 0.01$, ** $p < 0.05$ vs. vehicle + sham animals; # $p < 0.05$ vs. vehicle + MCAO animals.

expression in MCAO damage was 0.41 ± 0.03 in vehicle animals and 0.92 ± 0.08 in baicalin animals (**Fig. 4B**). In MCAO damage, the level of Bax expression was 1.38 ± 0.07 and 1.09 ± 0.06 in vehicle- and baicalin-treated animals, respectively (**Fig. 4B**). The Bcl-2/Bax ratio levels in MCAO damage were 0.29 ± 0.04 in vehicle animals and 0.83 ± 0.05 in baicalin animals (**Fig. 4C**). Immunofluorescence staining showed that baicalin attenuated the changes in Bcl-2 and Bax caused by MCAO damage (**Fig. 4E and F**). After MCAO damage, the number of Bcl-2-positive cells in the cerebral cortex decreased and the number of Bax-positive cells increased. Bcl-2 levels in MCAO animals were 0.32 ± 0.04 and 0.93 ± 0.04 in vehicle- and baicalin-treated

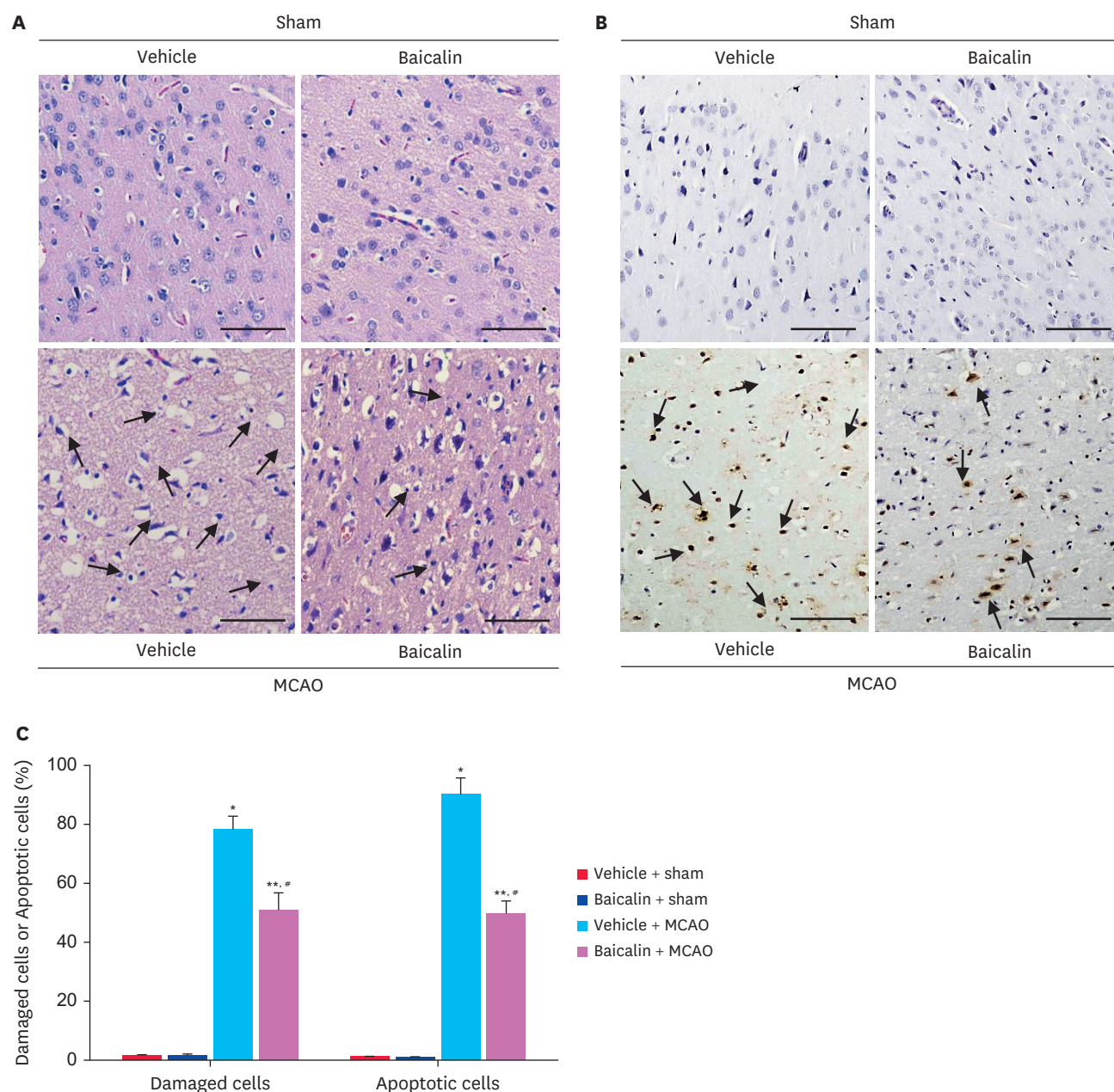


Fig. 3. Baicalin attenuates histopathological changes in MCAO animals. Representative microphotograph of (A) H&E staining and (B) TUNEL staining of the cerebral cortex in vehicle + sham, baicalin + sham, vehicle + MCAO, and baicalin + MCAO animals. (C) The number of damaged cells and TUNEL-positive cells increased in vehicle-treated + MCAO animals, whereas baicalin treatment attenuated these increases. Arrows indicate damaged cells with condensed and shrunken nuclei, swollen shape, and vacuolated forms (A). Arrows indicate the TUNEL-positive cells with dark brown (B). Data (n = 4) are presented as mean $\pm$ standard error of means. Scale bar: 100 μ m.

H&E, hematoxylin and eosin; TUNEL, terminal deoxynucleotidyl transferase dUTP nick end labeling; MCAO, middle cerebral artery occlusion.

* $p < 0.01$, ** $p < 0.05$ vs. vehicle + sham animals; # $p < 0.05$ vs. vehicle + MCAO animals.

animals, respectively (**Fig. 4D**). Bax levels in MCAO animals were 1.34 ± 0.05 and 1.08 ± 0.05 in vehicle- and baicalin-treated animals, respectively (**Fig. 4D**). The Bcl-2/Bax ratio levels were 0.23 ± 0.04 in vehicle + MCAO and 0.85 ± 0.02 in baicalin + MCAO animals (**Fig. 4C**). Caspase-3 expression was elevated in MCAO animals, but baicalin treatment prevented this increase (**Fig. 5A**). Caspase-3 levels were 3.17 ± 0.17 and 1.63 ± 0.14 in vehicle- and baicalin-treated MCAO animals (**Fig. 5B**). Caspase-3 expression in brain tissues was further confirmed

by immunofluorescence staining (**Fig. 5C**). The number of caspase-3 positive cells was higher in vehicle-treated MCAO animals compared to baicalin-treated animals. Caspase-3 levels in MCAO animals were 3.45 ± 0.34 in vehicle animals and 1.58 ± 0.22 in baicalin animals (**Fig. 5D**).

DISCUSSION

Ischemic stroke leads to cognitive impairment and hemiplegia, which leads to irreversible neurological disorders [18,19]. We clearly confirmed that MCAO damage results in significant neurobehavioral deficits. MCAO surgery is a widely used animal model for studying stroke. We demonstrated that baicalin significantly reduces neuronal damage induced by MCAO, as evidenced by various neurological behavioral tests. Baicalin effectively ameliorated sensorimotor impairment and body imbalance caused by MCAO. A previous study also demonstrated the neuroprotective effects of baicalin on neurological deficit scores in rats [20]. We extended the scope of the analysis using a wider range of behavioral tests, including neurological deficit score test, corner test, vibrissae-induced forelimb placement test, and grip strength test. Our results clearly highlight the protective effects of baicalin by improving neurological behavioral disorders.

Oxidative stress is a critical factor in the pathophysiology of ischemic stroke [21]. MCAO induces blockage of blood flow, deprivation of oxygen and nutrients, and subsequent energy deficiency, leading to neuronal cell death [22]. Ischemic stroke not only increases oxidative stress and pyroptosis but also activates neuroinflammation in microglia [23]. It also increases superoxide dismutase activity [23]. Excessive production of ROS induces oxidative stress, leading to cellular dysfunction, including DNA damage and membrane lipid peroxidation, subsequently triggers caspase activation and apoptosis [24]. Baicalin inhibits ROS production and stabilizes glutamine synthase under oxidative stress induced by glutamate exposure, thereby reducing excitotoxicity [25]. In particular, the reduction of LPO by baicalin suggests that baicalin effectively preserves membrane integrity by decreasing oxidative damage to polyunsaturated fatty acids, which play a key role in neuronal cell membranes [26]. LPO is recognized as a critical mechanism of oxidative damage in neuronal cells, directly disrupting membrane integrity and generating various secondary reactive compounds that amplify neuronal injury [27]. MDA, a major reactive aldehyde generated during LPO, leads to structural alterations and functional impairments, thereby contributing to neuronal dysfunction and neurodegeneration due to the high susceptibility of neuronal membranes to oxidative damage [28]. Our results demonstrated that baicalin treatment mitigated ROS accumulation and LPO during MCAO damage. These findings support the fact that baicalin acts as a potent antioxidant in ischemic stroke models.

Cerebral ischemia induces irreversible brain damage through complex pathophysiological mechanisms [29]. It degrades tight junction proteins, compromises BBB integrity, and leads to brain edema and nerve damage [30]. Baicalin decreases matrix metalloproteinase-9 expression and decreases BBB permeability following MCAO, thereby reducing cerebral edema [31]. We showed that baicalin administration ameliorated infarction and histopathological damage caused by MCAO damage. In addition, TUNEL histochemical staining was conducted to evaluate apoptosis. MCAO animals showed an increased number of TUNEL-positive cells in the ipsilateral cerebral cortex, indicating that MCAO surgery induces apoptosis. However, the administration of baicalin alleviated this increase and reduced apoptosis by MCAO, demonstrating the neuroprotective effect of baicalin.

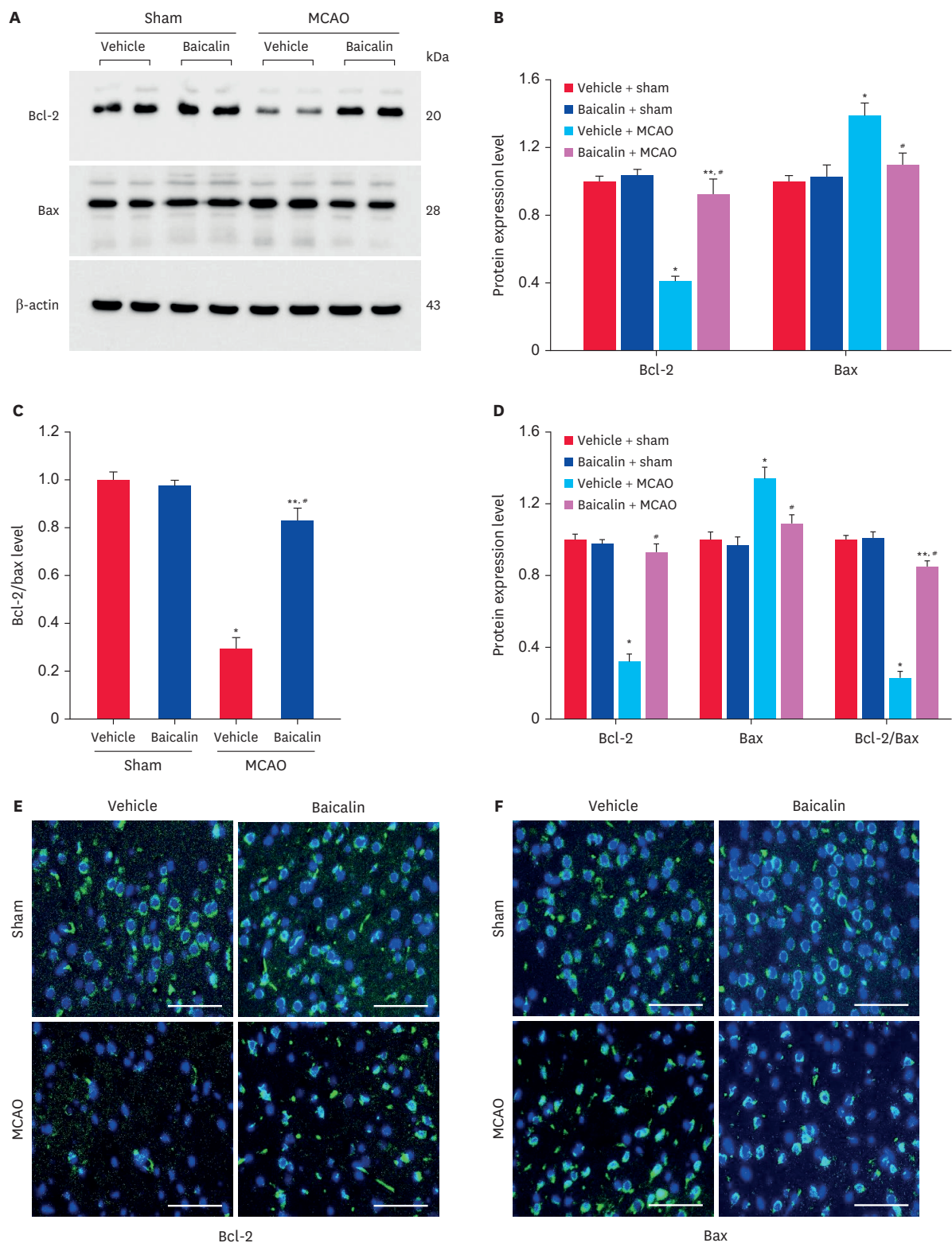


Fig. 4. Baicalin regulates Bcl-2 and Bax expressions in MCAO animals. (A-C) Western blot analysis and (D-F) immunofluorescence staining of Bcl-2 and Bax in the cerebral cortex of vehicle + sham, baicalin + sham animal, vehicle + MCAO, and baicalin + MCAO animals. Western blot analysis is presented as the ratio of each protein density to β -actin density. Results of immunostaining were expressed as a ratio of the number of each positive cell (green color) to that of DAPI-positive cells (blue color). The result for vehicle + sham animals was set to 1.0. Data (n = 4) are represented as mean $\pm$ standard error of means. Scale bar = 75 μ m. MCAO, middle cerebral artery occlusion; DAPI, 4',6-diamidino-2-phenylindole.

* p < 0.01, ** p < 0.05 vs. vehicle + sham animals, # p < 0.05 vs. vehicle + MCAO animals.

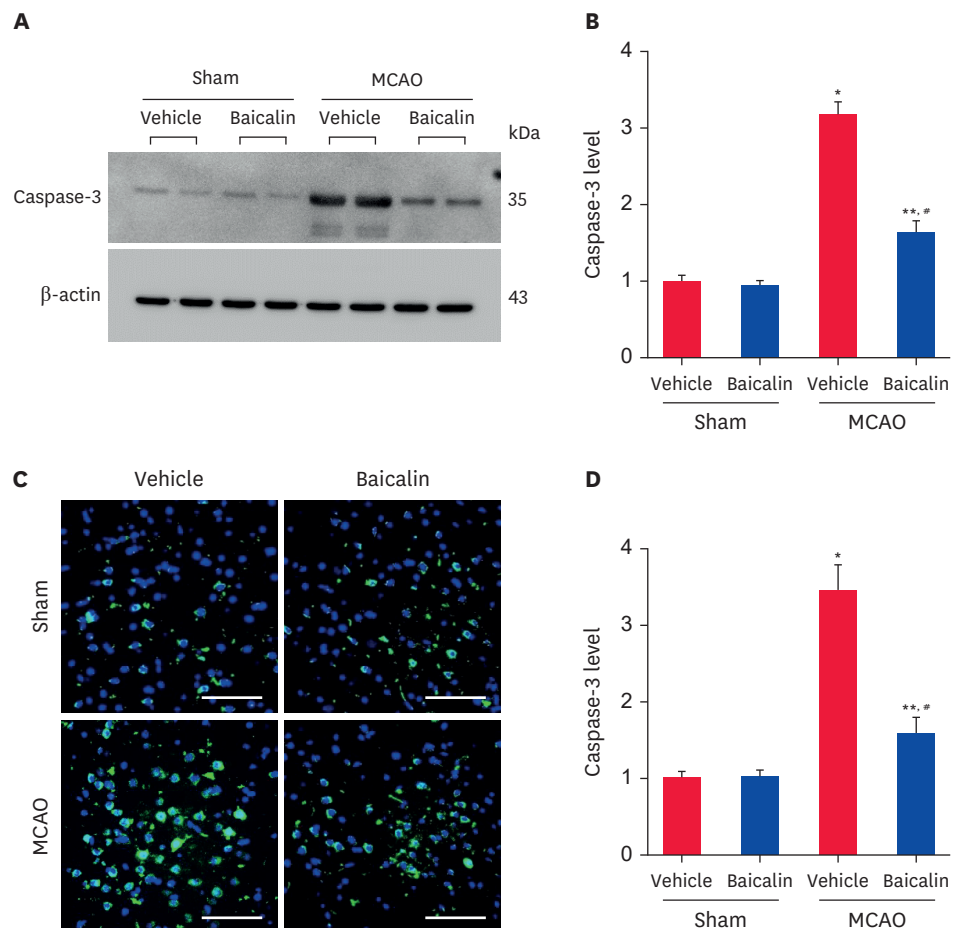


Fig. 5. Western blot analysis and immunofluorescence staining of caspase-3 in the cerebral cortex of vehicle + sham, baicalin + sham animal, vehicle + MCAO, and baicalin + MCAO animals. Western blot analysis are presented as the ratio of each protein density to β -actin density (A and B). Results of immunostaining were expressed as a ratio of the number of caspase-3 positive cells (green color) to that of DAPI positive cells (blue color) (C and D). The result for vehicle + sham animals was set to 1.0. Data (n = 4) are represented as mean $\pm$ standard error of means. Scale bar = 75 μ m.

MCAO, middle cerebral artery occlusion; DAPI, 4',6-diamidino-2-phenylindole.

* $p < 0.01$, ** $p < 0.05$ vs. vehicle + sham animals; # $p < 0.05$ vs. vehicle + MCAO animals.

Bcl-2 family proteins play a crucial role as regulators affecting cell fate [32]. They are involved in regulating apoptosis, some members promote cell survival and others promote apoptosis. This balance is important in maintaining cell homeostasis and responding to various stress signals. Among the Bcl-2 family proteins, Bcl-2 is a representative anti-apoptotic protein, and Bax is known as a pro-apoptotic protein [32,33]. Bcl-2 prevents apoptosis by inhibiting the process of apoptosis, whereas Bax promotes apoptosis by stimulating the release of cytochrome c from mitochondria, leading to activation of the apoptotic cascade [32-34]. Cerebral ischemia induced neuronal death by down- and up-regulation of Bcl-2 and Bax and activation of caspase activity [35,36]. Overexpression of Bcl-2 in stroke model prevents neuronal damage and preserves neuron from damage [37]. We found that baicalin treatment alleviated a decrease in Bcl-2 and an increase in Bax caused by MCAO surgery. The interaction between Bcl-2 and Bax, specifically the ratio of these proteins, is crucial for regulating cell survival and death [34]. An increase in the Bcl-2/Bax ratio inhibits apoptosis, while a decrease in this ratio promotes apoptosis. Our results showed that baicalin treatment during cerebral ischemia alleviated the decrease in Bcl-2/Bax ratio. These findings demonstrate that baicalin maintains a balance between anti-apoptotic

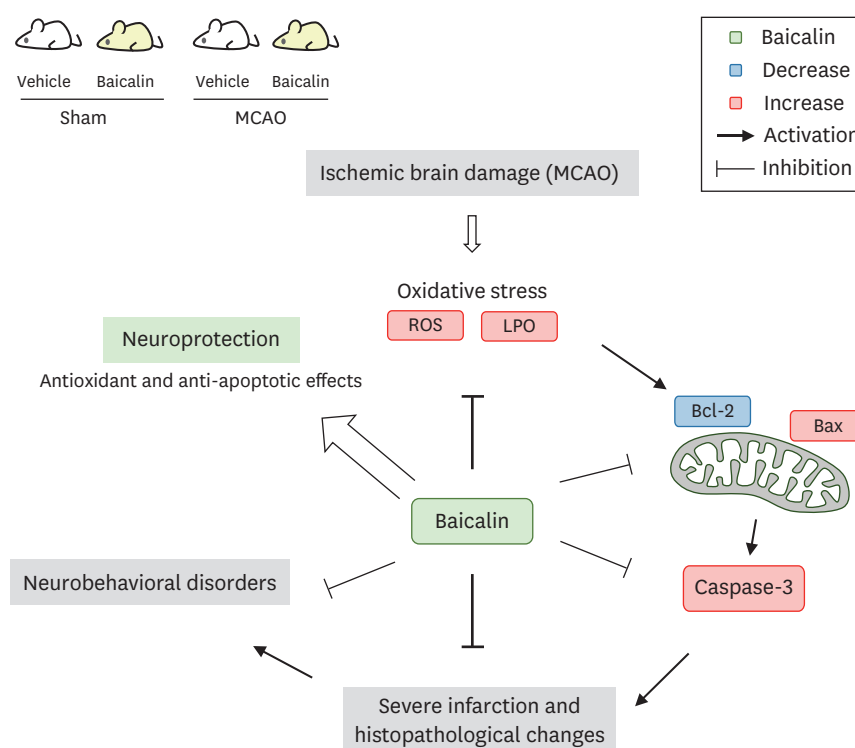


Fig. 6. A schematic of the neuroprotective effects of baicalin in focal cerebral ischemia. MCAO damage induced morphological changes, cerebral edema, infarction, and neurological disorders, and baicalin attenuated these changes through antioxidant and anti-apoptotic effects. MCAO damage decreased the expression of Bcl-2 and increased the expression of Bax and caspase-2, whereas baicalin alleviated these changes. MCAO, middle cerebral artery occlusion.

and pro-apoptotic proteins, potentially providing neuroprotective effects. Caspase-3 is a well-characterized apoptotic protein and is widely used as an indicator for apoptosis [38]. Our results confirmed the upregulation of caspase-3 in cerebral ischemia, and baicalin treatment prevented this increase in response to MCAO damage. Our results indicate that baicalin protects neurons from ischemic injury by inhibiting apoptosis. This suggests that baicalin plays an essential role in promoting the survival of neurons during ischemic injury. A recent study also reported that baicalin effectively mitigates apoptosis and inflammatory responses, while preserving tight junction integrity in both in vitro and in vivo models of ischemic stroke, further supporting its broad neurovascular protective effects [39]. This study focused on the protective effect of baicalin through various behavioral tests in the MCAO animal model and evaluated the neuroprotective effects of baicalin via antioxidant and anti-apoptotic pathways. However, further investigation into other mechanisms is necessary. Our future research will focus on baicalin's effects through various mechanisms, including glial cell activation, mitochondrial dysfunction, autophagy, and neurogenesis. Glial cells play an important role in recovery after nerve damage and ischemic stroke [40,41]. Based on this research, future studies will focus on demonstrating how baicalin regulates the dynamics of glial cells during the acute and recovery phases of ischemic brain injury. Our results provide evidence that baicalin acts as an effective therapeutic agent to alleviate ischemic brain injury. We schematized these results in **Fig. 6**. In conclusion, ischemic brain injury resulted in neurological disorders and histopathological changes, while baicalin alleviated these alterations through its antioxidant and anti-apoptotic effects. This suggests that baicalin can be a beneficial agent for mitigating the consequences of ischemic brain injury.

REFERENCES

1. Ingall T. Stroke--incidence, mortality, morbidity and risk. *J Insur Med.* 2004;36(2):143-152. [PUBMED](#)
2. Hankey GJ. Stroke. *Lancet.* 2017;389(10069):641-654. [PUBMED](#) | [CROSSREF](#)
3. Katan M, Luft A. Global burden of stroke. *Semin Neurol.* 2018;38(2):208-211. [PUBMED](#) | [CROSSREF](#)
4. Jurcau A, Simion A. Neuroinflammation in cerebral ischemia and ischemia/reperfusion injuries: from pathophysiology to therapeutic strategies. *Int J Mol Sci.* 2021;23(1):14. [PUBMED](#) | [CROSSREF](#)
5. Hou ST, MacManus JP. Molecular mechanisms of cerebral ischemia-induced neuronal death. *Int Rev Cytol.* 2002;221:93-148. [PUBMED](#) | [CROSSREF](#)
6. Mattson MP, Culmsee C, Yu ZF. Apoptotic and antiapoptotic mechanisms in stroke. *Cell Tissue Res.* 2000;301(1):173-187. [PUBMED](#) | [CROSSREF](#)
7. Kyo R, Nakahata N, Sakakibara I, Kubo M, Ohizumi Y. Baicalin and baicalein, constituents of an important medicinal plant, inhibit intracellular Ca^{2+} elevation by reducing phospholipase C activity in C6 rat glioma cells. *J Pharm Pharmacol.* 1998;50(10):1179-1182. [PUBMED](#) | [CROSSREF](#)
8. Cao Y, Mao X, Sun C, Zheng P, Gao J, Wang X, et al. Baicalin attenuates global cerebral ischemia/reperfusion injury in gerbils via anti-oxidative and anti-apoptotic pathways. *Brain Res Bull.* 2011;85(6):396-402. [PUBMED](#) | [CROSSREF](#)
9. Cheng F, Lu Y, Zhong X, Song W, Wang X, Sun X, et al. Baicalin's therapeutic time window of neuroprotection during transient focal cerebral ischemia and its antioxidative effects *in vitro* and *in vivo*. *Evid Based Complement Alternat Med.* 2013;2013:120261. [PUBMED](#) | [CROSSREF](#)
10. Singh S, Meena A, Luqman S. Baicalin mediated regulation of key signaling pathways in cancer. *Pharmacol Res.* 2021;164:105387. [PUBMED](#) | [CROSSREF](#)
11. Ma Z, Feng D, Rui W, Wang Z. Baicalin attenuates chronic unpredictable mild stress-induced hippocampal neuronal apoptosis through regulating SIRT1/PARP1 signaling pathway. *Behav Brain Res.* 2023;441:114299. [PUBMED](#) | [CROSSREF](#)
12. Li Y, Zhuang P, Shen B, Zhang Y, Shen J. Baicalin promotes neuronal differentiation of neural stem/progenitor cells through modulating p-stat3 and bHLH family protein expression. *Brain Res.* 2012;1429:36-42. [PUBMED](#) | [CROSSREF](#)
13. Ge QF, Hu X, Ma ZQ, Liu JR, Zhang WP, Chen Z, et al. Baicalin attenuates oxygen-glucose deprivation-induced injury via inhibiting NMDA receptor-mediated 5-lipoxygenase activation in rat cortical neurons. *Pharmacol Res.* 2007;55(2):148-157. [PUBMED](#) | [CROSSREF](#)
14. Tu XK, Yang WZ, Shi SS, Wang CH, Chen CM. Neuroprotective effect of baicalin in a rat model of permanent focal cerebral ischemia. *Neurochem Res.* 2009;34(9):1626-1634. [PUBMED](#) | [CROSSREF](#)
15. Longa EZ, Weinstein PR, Carlson S, Cummins R. Reversible middle cerebral artery occlusion without craniectomy in rats. *Stroke.* 1989;20(1):84-91. [PUBMED](#) | [CROSSREF](#)
16. Woodlee MT, Asseo-García AM, Zhao X, Liu SJ, Jones TA, Schallert T. Testing forelimb placing "across the midline" reveals distinct, lesion-dependent patterns of recovery in rats. *Exp Neurol.* 2005;191(2):310-317. [PUBMED](#) | [CROSSREF](#)
17. Esterbauer H, Schaur RJ, Zollner H. Chemistry and biochemistry of 4-hydroxynonenal, malonaldehyde and related aldehydes. *Free Radic Biol Med.* 1991;11(1):81-128. [PUBMED](#) | [CROSSREF](#)
18. Piechowski-Jozwiak B, Bogousslavsky J. Neurobehavioral syndromes. *Front Neurol Neurosci.* 2012;30:57-60. [PUBMED](#) | [CROSSREF](#)
19. Park DJ, Kang JB, Shah FA, Koh PO. Resveratrol modulates the Akt/GSK-3 β signaling pathway in a middle cerebral artery occlusion animal model. *Lab Anim Res.* 2019;35(1):18. [PUBMED](#) | [CROSSREF](#)
20. Xue X, Qu XJ, Yang Y, Sheng XH, Cheng F, Jiang EN, et al. Baicalin attenuates focal cerebral ischemic reperfusion injury through inhibition of nuclear factor κ B p65 activation. *Biochem Biophys Res Commun.* 2010;403(3-4):398-404. [PUBMED](#) | [CROSSREF](#)
21. Darvesh AS, Carroll RT, Bishayee A, Geldenhuys WJ, Van der Schyf CJ. Oxidative stress and Alzheimer's disease: dietary polyphenols as potential therapeutic agents. *Expert Rev Neurother.* 2010;10(5):729-745. [PUBMED](#) | [CROSSREF](#)
22. Cespedes A, Villa M, Benito-Cuesta I, Perez-Alvarez MJ, Ordoñez L, Wandosell F. Energy-sensing pathways in ischemia: the counterbalance between AMPK and mTORC. *Curr Pharm Des.* 2019;25(45):4763-4770. [PUBMED](#) | [CROSSREF](#)
23. Liang YB, Song PP, Zhu YH, Xu JM, Zhu PZ, Liu RR, et al. TREM-1-targeting LP17 attenuates cerebral ischemia-induced neuronal injury by inhibiting oxidative stress and pyroptosis. *Biochem Biophys Res Commun.* 2020;529(3):554-561. [PUBMED](#) | [CROSSREF](#)

24. Hwang IK, Yoo KY, Kim DW, Lee CH, Choi JH, Kwon YG, et al. Changes in the expression of mitochondrial peroxiredoxin and thioredoxin in neurons and glia and their protective effects in experimental cerebral ischemic damage. *Free Radic Biol Med*. 2010;48(9):1242-1251. [PUBMED](#) | [CROSSREF](#)
25. Song X, Gong Z, Liu K, Kou J, Liu B, Liu K. Baicalin combats glutamate excitotoxicity via protecting glutamine synthetase from ROS-induced 20S proteasomal degradation. *Redox Biol*. 2020;34:101559. [PUBMED](#) | [CROSSREF](#)
26. Liao PR, Wu MS, Lee CK. Inhibitory effects of *Scutellaria baicalensis* root extract on linoleic acid hydroperoxide-induced lung mitochondrial lipid peroxidation and antioxidant activities. *Molecules*. 2019;24(11):2143. [PUBMED](#) | [CROSSREF](#)
27. Sultana R, Perluigi M, Butterfield DA. Lipid peroxidation triggers neurodegeneration: a redox proteomics view into the Alzheimer disease brain. *Free Radic Biol Med*. 2013;62:157-169. [PUBMED](#) | [CROSSREF](#)
28. Reed TT. Lipid peroxidation and neurodegenerative disease. *Free Radic Biol Med*. 2011;51(7):1302-1319. [PUBMED](#) | [CROSSREF](#)
29. Durukan A, Tatlisumak T. Acute ischemic stroke: overview of major experimental rodent models, pathophysiology, and therapy of focal cerebral ischemia. *Pharmacol Biochem Behav*. 2007;87(1):179-197. [PUBMED](#) | [CROSSREF](#)
30. Imai T, Takagi T, Kitashoji A, Yamauchi K, Shimazawa M, Hara H. Nrf2 activator ameliorates hemorrhagic transformation in focal cerebral ischemia under warfarin anticoagulation. *Neurobiol Dis*. 2016;89:136-146. [PUBMED](#) | [CROSSREF](#)
31. Tu XK, Yang WZ, Liang RS, Shi SS, Chen JP, Chen CM, et al. Effect of baicalin on matrix metalloproteinase-9 expression and blood-brain barrier permeability following focal cerebral ischemia in rats. *Neurochem Res*. 2011;36(11):2022-2028. [PUBMED](#) | [CROSSREF](#)
32. Hardwick JM, Soane L. Multiple functions of Bcl-2 family proteins. *Cold Spring Harb Perspect Biol*. 2013;5(2):a008722. [PUBMED](#) | [CROSSREF](#)
33. Oltvai ZN, Millman CL, Korsmeyer SJ. Bcl-2 heterodimerizes *in vivo* with a conserved homolog, Bax, that accelerates programmed cell death. *Cell*. 1993;74(4):609-619. [PUBMED](#) | [CROSSREF](#)
34. Korsmeyer SJ, Shutter JR, Veis DJ, Merry DE, Oltvai ZN. Bcl-2/Bax: a rheostat that regulates an anti-oxidant pathway and cell death. *Semin Cancer Biol*. 1993;4(6):327-332. [PUBMED](#)
35. Li T, Su D, Lu H, Gao Y, Liu Y, Wang S, et al. Recombinant human brain natriuretic peptide attenuates ischemic brain injury in mice by inhibiting oxidative stress and cell apoptosis via activation of PI3K/AKT/Nrf2/HO-1 pathway. *Exp Brain Res*. 2023;241(11-12):2751-2763. [PUBMED](#) | [CROSSREF](#)
36. Tanbek K, Yuksel F, Tekin S, Tekin C, Sandal S. Asprosin improved neuronal survival by suppressing apoptosis and enhancing the activity of the autophagy pathway in the MCAO model in rats. *Eur Rev Med Pharmacol Sci*. 2024;28(5):1937-1946. [PUBMED](#) | [CROSSREF](#)
37. Zhao H, Yenari MA, Cheng D, Barreto-Chang OL, Sapolsky RM, Steinberg GK. Bcl-2 transfection via herpes simplex virus blocks apoptosis-inducing factor translocation after focal ischemia in the rat. *J Cereb Blood Flow Metab*. 2004;24(6):681-692. [PUBMED](#) | [CROSSREF](#)
38. Salvesen GS. Caspases and apoptosis. *Essays Biochem*. 2002;38:9-19. [PUBMED](#) | [CROSSREF](#)
39. Hao Z, Zhang Z, Zhao Y, Wang D. Baicalin reduces immune cell infiltration by inhibiting inflammation and protecting tight junctions in ischemic stroke injury. *Am J Chin Med*. 2023;51(2):355-372. [PUBMED](#) | [CROSSREF](#)
40. Kim JY, Park J, Chang JY, Kim SH, Lee JE. Inflammation after ischemic stroke: the role of leukocytes and glial cells. *Exp Neurobiol*. 2016;25(5):241-251. [PUBMED](#) | [CROSSREF](#)
41. Nedergaard M, Dirnagl U. Role of glial cells in cerebral ischemia. *Glia*. 2005;50(4):281-286. [PUBMED](#) | [CROSSREF](#)